

# How to Update Existing Physics Pipelines to Test Unified Substrate Theory

A Technical Implementation Guide for Research Teams

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## Abstract

Modern astrophysics, cosmology, and field-theory pipelines already contain all the machinery required to test Unified Substrate Theory. No new detectors, frameworks, or solvers are needed. The required changes are small, local modifications to existing code: replacing ontology assumptions, swapping a Lagrangian, redefining constants, and adding a few derived quantities. This document provides explicit instructions for updating gamma-ray burst (GRB) dispersion pipelines, gravitational-wave analysis tools, dark-matter reconstruction codes, cosmological expansion models, and field-theory solvers to evaluate UST predictions. Every step is implementable in a single workday using current development tools.

## 1. Introduction

Unified Substrate Theory replaces the standard ontology of “particles + fields + spacetime” with a single complex substrate field  $\Phi$  whose density  $\rho_s$  and phase  $\theta$  generate all physical behavior. In UST, particles are localized knots of substrate motion, fields are patterns of phase and density, and spacetime geometry emerges from the stiffness and distribution of the substrate. Because UST is expressed through a compact action and simple derived quantities, existing physics pipelines can be adapted with minimal code changes. This document outlines those changes explicitly, focusing on practical implementation steps for research teams.

## 2. Updating Field-Theory Models

Most physics engines define a Lagrangian or action at the top of the model. To test UST, the standard field stack (electromagnetism, general relativity, and quantum field theory) is replaced with the substrate action:

### Code

$$S[\Phi] = \int d^4x \left[ \frac{1}{2} K_s |\partial_\mu \Phi|^2 - V(\rho_s) \right]$$

Here,  $\Phi$  is a complex scalar field,  $K_s$  is the substrate stiffness, and  $V(\rho_s)$  is a potential depending on the substrate density  $\rho_s$ .

### Required code changes

- Replace the existing Lagrangian with the substrate action  $S[\Phi]$ .
- Define  $\Phi$  as a complex scalar field.
- Define  $\rho_s = |\Phi|^2$  and  $\theta = \arg(\Phi)$ .
- Remove explicit particle fields; particles become localized knot solutions of  $\Phi$ .
- Update solvers to track phase gradients instead of separate electric and magnetic fields.

This is a drop-in replacement for the existing field definition and can be implemented without changing the underlying numerical solvers.

### 3. Updating Charge, Spin, and Electromagnetic Behavior

In UST, charge, spin, and electromagnetic behavior are derived from the phase and topology of the substrate field.

Charge becomes phase winding:

#### Code

$$q = (1/2\pi) \oint \nabla \theta \cdot d\mathbf{l}$$

Spin becomes a topological linking integral:

#### Code

$$S = \int \mathbf{A} \cdot \mathbf{B} \, d^3x$$

Electromagnetism becomes phase-gradient behavior:

#### Code

$$A_\mu = \partial_\mu \theta$$

### Required code changes

- Replace charge as a fixed parameter with charge defined as a winding number of the phase  $\theta$ .
- Replace spin as an intrinsic quantum number with spin defined via the linking integral  $S$ .
- Replace electromagnetic fields with phase-gradient fields derived from  $A_\mu = \partial_\mu \theta$ .

These are function-definition changes, not new solvers, and can be implemented by updating the model's derived quantities.

#### 4. Updating Constants (c and G)

In UST, the speed of light and gravitational strength are not fundamental inputs but emergent quantities determined by substrate properties.

The speed of light becomes:

##### Code

$$c^2 = K_s / \rho_s$$

Gravitational strength becomes:

##### Code

$$G = g(K_s)$$

##### Required code changes

- Replace hard-coded c with a computed value from the ratio of stiffness  $K_s$  to density  $\rho_s$ .
- Replace G with a function of substrate stiffness  $g(K_s)$ .

These are constant-definition changes, not structural changes to the models.

#### 5. Updating GRB Photon-Dispersion Pipelines

Existing GRB pipelines already compute arrival-time dispersion as a function of photon energy and source properties. To test UST, these pipelines are modified to incorporate substrate-dependent propagation speed.

##### Required code changes

- Add energy-dependent propagation speed using  $c(\rho_s)$ , where  $c^2 = K_s / \rho_s$ .
- Fit arrival-time lags to substrate-granularity dispersion curves predicted by UST.
- Subtract astrophysical noise and source-intrinsic effects as in standard analyses.

This is a parameter-update, not a new pipeline, and can be implemented by modifying the propagation model in existing GRB timing codes.

## 6. Updating Gravitational-Wave Dispersion Pipelines

Current gravitational-wave (GW) pipelines already split waveforms by frequency and analyze arrival times and phase evolution. To test UST, these tools are extended to allow frequency-dependent propagation speed.

### Required code changes

- Add frequency-dependent propagation speed derived from substrate stiffness  $K_s$ .
- Fit arrival-time offsets and phase shifts to UST dispersion curves.
- Compare results to the general-relativistic baseline of zero dispersion.

This is a single additional function in the waveform propagation model and does not require rewriting the GW analysis framework.

## 7. Updating Dark-Matter Reconstruction Pipelines

Rotation-curve and lensing pipelines already reconstruct gravitational potentials and infer dark-matter distributions. In UST, dark matter corresponds to elevated substrate density without knot structures.

### Required code changes

- Convert gravitational potential fields into substrate density gradients.
- Identify regions of elevated density that do not correspond to localized knot solutions (non-particle substrate).
- Predict rotation curves and lensing behavior from substrate density alone.

This is a thresholding rule change and a reinterpretation of the reconstructed fields, not a new model.

## 8. Updating Dark-Energy Cosmology Pipelines

Cosmology engines already compute baryon acoustic oscillations (BAO), supernova (SN Ia) distances, and expansion histories. In UST, dark energy is modeled as a large-scale pressure arising from substrate elasticity.

### Required code changes

- Compute density deviations from equilibrium substrate configuration.
- Apply substrate elasticity to compute restoring pressure associated with these deviations.
- Predict expansion acceleration from the resulting pressure term.

This is a pressure-term update within existing cosmological models and does not require a new cosmology framework.

## **9. Updating Fine-Structure Constant Solvers**

Geometry-optimization engines and field-theory solvers already exist for exploring stable configurations. In UST, the fine-structure constant is targeted as a geometric property of minimal electromagnetic knots in the substrate.

### **Required code changes**

- Model minimal electromagnetic knot geometry in substrate units.
- Solve for stable loop configurations of  $\Phi$  that correspond to electromagnetic knots.
- Compute the resulting dimensionless coupling ratio.
- Check convergence near the observed value of approximately  $1/137$ .

This is a geometry-solver task, not a new theory engine, and can be implemented using existing optimization and field-solver tools.

## **10. Conclusion**

Testing Unified Substrate Theory does not require new physics infrastructure. It requires small, local code updates to existing pipelines. By replacing the standard field stack with the substrate action, redefining charge, spin, and electromagnetic behavior in terms of phase and topology, and updating constant definitions and propagation models, research teams can evaluate all major UST predictions using current tools. Modern AI-assisted development environments make these changes implementable in a single workday. The barrier to testing UST is not technical; it is conceptual.